Multiple Linear Regression

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Introduction

Multiple linear regression extends simple regression to include two or more predictors. Each coefficient represents the partial effect of its predictor, holding all others constant. This allows confounding adjustment, multi-factor prediction, and partial correlations.

Prerequisites

Simple linear regression.

Theory

Model: $Y = \beta_0 + \beta_1 X_1 + \ldots + \beta_p X_p + \varepsilon$, with $\varepsilon \sim \mathcal{N}(0, \sigma^2)$.

OLS estimates: $\hat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top Y$. Each coefficient is the **partial slope**: change in $E[Y]$ per unit of X_j, holding other predictors fixed.

Collinearity: highly correlated predictors inflate SE of individual coefficients. Diagnose with variance inflation factors (VIF): $\text{VIF}_j = 1/(1 - R_j^2)$, where R_j^2 is the R^2 of regressing X_j on the others. VIF > 5 is concerning; > 10 is severe.

R^2 is the proportion of variance explained; adjusted R^2 penalises model complexity.

Assumptions

Linearity of each predictor, independence, homoscedasticity, normality of residuals, no severe collinearity.

R Implementation

```r
library(broom); library(car); library(performance)

data(mtcars)
fit <- lm(mpg ~ wt + hp + cyl + disp, data = mtcars)

tidy(fit, conf.int = TRUE)
glance(fit)
vif(fit)

check_model(fit)
```

Output & Results

Coefficients with SE, t-statistic, p-values, 95 % CIs. R^2, adjusted R^2, residual SE. VIF values per predictor.

Interpretation

"After adjusting for horsepower, cylinders, and displacement, each additional 1000 lbs of weight reduced mpg by 3.8 (95 % CI -6.8 to -0.7). The model explained 84 % of the variance in mpg ($R^2 = 0.84$)."

Practical Tips

- Centre continuous predictors for interpretable intercepts.
- For highly collinear predictors, drop one, combine them, or use ridge regression.
- Always check residual plots before trusting p-values.
- Report both raw coefficients and standardised ones when comparing effect magnitudes across predictors on different scales.
- For prediction accuracy, cross-validate – in-sample R^2 is optimistic.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Correlation vs regression; Model-I/II regression
- Simple linear regression
- Multiple regression: confounding, interaction, centring
- Diagnostics: residuals, QQ, leverage, Cook's distance, VIF
- Robust regression, weighted LS, HC (sandwich) standard errors
- One-way ANOVA and contrasts (emmeans)
- Two-way / factorial ANOVA with interaction
- RCBD and repeated measures → mixed models
- GAMs with mgcv
- Non-linear regression with nls
- Logistic regression (binomial GLM)
- ANCOVA in RCTs (baseline adjustment)
- Ordinal and multinomial regression
- Poisson and negative-binomial regression
- Calibration, discrimination, ROC/AUC, Brier score
- Dichotomisation, change scores, regression to the mean
- Decision curves, NRI, IDI